

Day 18 - Lecture 18.9: Vertical Bar Charts (`plt.bar`)

1. Overview & Definition

A **Bar Chart** represents categorical data with rectangular bars whose heights are proportional to the values they represent.

- **X-axis:** Discrete categories (e.g. Movie titles, product names, years).
- **Y-axis:** Numerical values or counts.

Difference Between Bar Chart and Histogram:

- **Bar Chart:** Visualizes categorical data. Bars have distinct spaces between them because categories are independent discrete entities.
 - **Histogram:** Visualizes continuous numerical intervals (bins). Bars touch each other.
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2. Syntax & Parameters

```
plt.bar(  
    x,                # Sequence of categories (strings, numbers, dates)  
    height,           # Numerical values representing bar heights  
    width=0.8,        # Bar width (default 0.8)  
    bottom=None,      # Baseline y-coordinate (used for stacked bars)  
    color='purple',   # Bar fill color  
    edgecolor='black', # Border color  
    linewidth=1.2     # Border thickness  
)
```

3. Code Implementation: Movie Box Office Revenues

```
import matplotlib.pyplot as plt

movies = [
    "The Dark Knight",
    "The Hurt Locker",
    "The King's Speech",
    "The Artist",
    "Argo"
]
years = [2008, 2009, 2010, 2011, 2012]
revenue = [1005, 170, 427, 133, 232] # in $M

plt.figure(figsize=(8, 5))
plt.bar(years, revenue, color="purple", edgecolor="black", width=0.6)

plt.title("Yearly Revenue of Oscar Movies", fontsize=14, fontweight="bold")
plt.xlabel("Years", fontsize=12)
plt.ylabel("Revenue (in $M)", fontsize=12)
plt.grid(axis='y', linestyle='--', alpha=0.6)
plt.show()
```

4. Key Takeaways

- Use `width` to control the thickness of bars (e.g., `width=0.5` makes them narrower).
- Adding `grid(axis='y')` provides horizontal reference lines without cluttering the vertical space.